

**Supermassive black hole**

At the center galaxy NGC 4258 is a vast black hole into which matter is spiraling, at the same time producing powerful jets of high-energy particles. These jets strike the disk of the galaxy and heat the gas there to thousands of degrees. That is why the center of the galaxy looks bright, not black. The image combines various types of radiation, including visible light (yellow), infrared (red), and X-rays (blue).